

Om Pranab Mohanty

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Undergraduate in Mathematical Sciences & Computing — Aspiring systems researcher with interests in Computer Architecture, Operating Systems, Performance Engineering, and Mathematical Modeling, seeking to build efficient, scalable, and high-performance computing systems through research and engineering.

EDUCATION

Sri Sathya Sai Institute of Higher Learning — Puttaparthi, India Jul 2023 – Present
B.S. (Hons. with Research) in Mathematical Sciences and Computing
Specialization: Artificial Intelligence — CGPA: 8.4 / 10 (O Grade) Rank 1 in cohort

Sri Sathya Sai Higher Secondary School — Puttaparthi, India Jul 2021 – Apr 2023
Intermediate Education (Class XII)
Subjects: Mathematics, Physics, Chemistry, Computer Science, English — Score: 85 / 100

WORK EXPERIENCE

Research Intern — Manipal Institute of Virology Mar 2026 – Present

- Worked on VAE-based modeling of long-term genomic data for dengue serotype analysis
- Focused on scalable representation learning, computational efficiency, and long-term genomic sequence modeling.

PROJECTS

LowBit: Lightweight Benchmarking and Instrumentation Toolkit [Project] Jul 2026 – Present

- Developed an open-source suite of deterministic microbenchmarks focused on computer architecture and machine learning kernel performance evaluation on RISC-V and released 2 benchmarks:
 - **LB-ML-001 (ml-transformer-attn_r)**: Implemented scalable multi-head attention with FP32 reference and symmetric INT8 quantization paths, including reproducible JSON workloads, golden-reference validation, cross-platform (native + RISC-V + QEMU) support, and microarchitectural analysis.
 - **LB-ML-002 (ml-gemm_r)**: Built dense GEMM kernel with naive, blocked-tiled, and SIMD-hinted variants, along with workload-driven testing, relative error validation, and arithmetic intensity analysis.

OS Synchronization as Symmetric Restriction in S_n [Code] Feb 2026 – Present

- Modeled OS synchronization (mutex, scheduling, deadlock) using group-theoretic structures in S_n
- Established safety–fairness trade-offs and introduced a metric for scheduling transitions
- Implemented validation framework in Python for small-scale systems

Network Ingress Subsystem in C [Code] Mar 2026 – Apr 2026

- Built a deterministic packet processing system using fixed-size buffers and protocol parsers
- Ensured predictable execution by eliminating dynamic memory allocation
- Validated behavior under constrained system conditions

SKILLS

Systems & Architecture — Computer Architecture, Operating Systems, Computer Networks,
Programming — C/C++, Python, SQL, Bash
Mathematics — Discrete Mathematics, Linear Algebra, Optimization, Probability
Tools — Git, Linux

COURSEWORK

Systems — Operating Systems, Computer Architecture, Computer Networks
Machine Learning — Data Mining and Machine Learning, Deep Learning, Natural Language Processing
Mathematics — Linear Algebra, Probability, Optimization, Discrete Mathematics, Abstract Algebra
Core CS — Algorithms

EXTRACURRICULAR ACTIVITIES

Secretary — Speech Craft Society, SSSIHL Feb 2025 – July 2026

- Led organization of student sessions on public speaking and structured communication